

Agent Architecture with Tool Use Report

Design

The design doesn't deviate much from the template, with only two notable changes:

1. We used an inherited method to generate tool schemas to standardize the way which those schemas are presented and stringified. This reduces the likelihood of inconsistencies as tool employment grows and reduces the chances of runtime surprises.
2. Tool calling is built around the Gemini API expectations, in lieu of plain prompt stuffing. This trades some more structured overhead and redtape for smoother and more reliable tool calling.

Test cases

Travel Policy

```
[jason@: % python app_starter.py "What do I need to know about the company's policy as it pertains to travel?"
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: What do I need to know about the company's policy as it pertains to travel?
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: policy_search
INFO:__main__:Mapped to internal tool policy_search (Search policy documents by keyword or topic)
INFO:__main__:Calling with arguments: {'query': 'travel'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
Answer
-----
: As an engineer, here's what you need to know about the company's travel policy:

**Pre-Approval is Required:**
All business travel must be pre-approved by your manager.

**Travel Budget Authorization (for Individual Contributors - ICs):**

*   **Domestic Travel:**
    *   **IC1-IC2:** $3,000 per trip limit, $15,000 per year limit
    *   **IC3-IC4:** $5,000 per trip limit, $25,000 per year limit
    *   **IC5+:** $10,000 per trip limit, no annual limit
*   **International Travel:**
    *   Requires VP approval.
    *   Budget limits are 50% higher than the domestic travel limits for your respective IC level.

Make sure to align your travel plans with these guidelines and get the necessary approvals before making any arrangements.

INFO:__main__:Tokens: 1747
INFO:__main__:Cost: $0.001483
INFO:__main__:Metrics: {'total_queries': 2, 'total_tokens': 1747, 'total_cost': 0.001483000000000001, 'avg_cost_per_query': 0.000741650000000001}
```

Layoffs

```
[jason@: % python app_starter.py "I'm managing a team of engineers but just started here. Are there any recent force reductions or layoffs I should be aware of?"
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: I'm managing a team of engineers but just started here. Are there any recent force reductions or layoffs I should be aware of?
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: policy_search
INFO:__main__:Mapped to internal tool policy_search (Search policy documents by keyword or topic)
INFO:__main__:Calling with arguments: {'query': 'layoffs'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
Answer
-----
: Based on the policy documents, there have been several discussions regarding layoffs, with the most recent meeting minutes dated March 18, 2026, and March 23, 2026.

However, the details within these documents regarding key decisions and action items are not clear and appear to be uninformative placeholder text. Therefore, I cannot provide specific details about the nature or scope of any force reductions from these records.

INFO:__main__:Tokens: 1668
INFO:__main__:Cost: $0.001321
INFO:__main__:Metrics: {'total_queries': 2, 'total_tokens': 1668, 'total_cost': 0.001321000000000001, 'avg_cost_per_query': 0.000660500000000001}
```

Expense Limits

```
[jason@: % python app_starter.py "I'm the VP of sales, what's my expense limit? "
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: I'm the VP of sales, what's my expense limit?
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: expense_query
INFO:__main__:Mapped to internal tool expense_query (Query expense approval limits by role)
INFO:__main__:Calling with arguments: {'role': 'vp'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
Answer
-----
: As the VP of Sales, your expense approval limit is $100,000.

INFO:__main__:Tokens: 980
INFO:__main__:Cost: $0.001497
INFO:__main__:Metrics: {'total_queries': 2, 'total_tokens': 980, 'total_cost': 0.0014973999999999999, 'avg_cost_per_query': 0.0007486999999999999}
```

Angela's Sales

```
[jason@: % python app_starter.py "I am responsible for this quarter's financial reporting but Angela is out. What meaningful sales did she pull in? I think this ]
might've been logged in some of the meeting minutes, check for hits on 'angela'"
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: I am responsible for this quarter's financial reporting but Angela is out. What meaningful sales did she pull in? I think this m
ight've been logged in some of the meeting minutes, check for hits on 'angela'
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: policy_search
INFO:__main__:Mapped to internal tool policy_search (Search policy documents by keyword or topic)
INFO:__main__:Calling with arguments: {'query': 'angela'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
Answer
-----
: It looks like you're asking about Angela Jones's sales figures, as "Angela Jones" is listed in the "Sales Territory 4 Guidelines" with a quota and specific ke
y accounts.

Here are the key accounts and their annual values associated with Angela Jones:

* **Quota:** $1,882,109
* **Key Accounts:**
  * Williams, Lopez and Martin: $51,420/year
  * Burnett Group: $20,912/year
  * Medina LLC: $64,823/year
  * Marquez-Holloway: $84,441/year
  * Nunez, Flowers and Rhodes: $72,121/year

Please note that another Angela, "Angela Orr," was mentioned in the "Q1 Planning - September Meeting" minutes under action items, but there were no sales figure
s or specific sales-related details associated with her.

INFO:__main__:Tokens: 1629
INFO:__main__:Cost: $0.002121
INFO:__main__:Metrics: {'total_queries': 2, 'total_tokens': 1629, 'total_cost': 0.0021211, 'avg_cost_per_query': 0.00106055}
```

The Brians

```
[jason@: % python app_starter.py "Do we have any employees named Brian?"
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: Do we have any employees named Brian?
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: employee_lookup
INFO:__main__:Mapped to internal tool employee_lookup (Find employee information by name or ID)
INFO:__main__:Calling with arguments: {'employee_name': 'Brian'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
Answer
```

: Yes, we have several employees named Brian:

```
* Brian Yang (Engineering)
* Brian Graham (Sales)
* Brian Anderson (Operations)
* Brian Johnson (Operations)
* Brian Hunt (Product)
* Brian Campbell (Engineering)
* Brian Hudson (Operations)
* Brian Vasquez (Finance)
* Brian Carrillo (Engineering)
* Brian Crosby (Marketing)
* Brian Smith (Sales)
* Brian Welch (Finance)
* Brian Cohen (Sales)
* Brian Collins (Sales)
* Brian Vance (Engineering)
* Brian Thornton (Engineering)
* Brianna Schroeder (Product)
* Brian Jacobson (Engineering)
* Brian Mckenzie (Product)
* Brian Rodriguez (Product)
* Brian Wilson (Engineering)
* Brian Cole (Sales)
* Brian Perez (Marketing)
* Brianna Williams (Finance)
* Brian Oneal (Human Resources)
* Brian Dillon (Human Resources)
* Brian Moses (Finance)
* Brian Herring (Engineering)
* Brian Harper (Operations)
* Briana Rodriguez (Sales)
* Brian Ellis (Finance)
* Brian Bowman (Sales)
* Brian Nguyen (Human Resources)
* Brian Jackson (Finance)
* Brian Allen (Engineering)
* Brian Fleming (Marketing)
* Brian Carter (Finance)
* Brian Baird (Product)
* Briana Juarez (Human Resources)
* Dr. Brianna Turner MD (Product)
* Brian Lane (Operations)
* Brian Griffin (Finance)
* Brian Huffman (Marketing)
* Brian Thompson (Engineering)
* Brianna Holmes (Marketing)
* Brian Thompson (Marketing)
* Brian Glenn (Marketing)
* Brian Taylor (Product)
* Brian Noble (Marketing)
* Brian Knapp (Human Resources)
* Brianna Smith (Engineering)
* Brian Cuevas (Human Resources)
* Brian Miller (Operations)
* Brian Morrison (Marketing)
* Brianna Johnson (Operations)
* Brian Nguyen (Marketing)
* Brian Reid (Marketing)
* Brian Henson (Finance)
* Brian Marshall (Finance)
* Brian Huffman (Sales)
* Brian Jones (Engineering)
* Brian Hill (Engineering)
* Brian Cole (Operations)
* Brian Hartman (Marketing)
* Brian Hamilton (Product)
* Brian Olson (Product)
* Brian Brown (Marketing)
* Brianna Hall (Finance)
* Brian Moore (Operations)
* Brian Fuller (Finance)
* Brian James (Product)
* Brian Phillips (Marketing)
* Brian Price (Marketing)
* Brian Moreno (Marketing)
* Brian Williams (Sales)
* Brian McIntosh (Human Resources)
* Brian Trevino (Human Resources)
* Brianna Oneal (Finance)
* Brian Alvarez (Human Resources)
* Brianna Ford (Finance)
* Brian Duncan (Product)
* Brian Cooper (Finance)
* Brian Franklin (Operations)
* Brian Richards (Product)
* Brian Bowman (Finance)
* Brian Johnson (Engineering)
* Brianna Adams (Human Resources)
* Brian Nguyen (Operations)
* Brian Solis (Sales)
```

```

* Brian Andrews (Marketing)
* Brian Hill (Sales)
* Brian Graham (Sales)
* Brian Maxwell (Product)
* Brian Wolfe (Sales)
* Brian Keller (Finance)
* Brianna Maldonado (Product)
* Brian Wheeler (Engineering)
* Brian Hernandez (Human Resources)
* Brian Schultz (Operations)
* Brian Morrison (Marketing)
* Brian Payne (Operations)
* Brian Soto (Marketing)
* Brian Le (Human Resources)
* Brianna Bentley (Marketing)

INFO:__main__:Tokens: 20287
INFO:__main__:Cost: $0.009472
INFO:__main__:Metrics: {'total_queries': 2, 'total_tokens': 20287, 'total_cost': 0.0094719, 'avg_cost_per_query': 0.00473595}

```

Danielle's Benefits

```

[jason@: % python app_starter.py "I'm just moved over from another segment's HR and trying to compile some benefits numbers. Can you tell me if Dnaiel Miller has
an HMO benefits plan?"
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: I'm just moved over from another segment's HR and trying to compile some benefits numbers. Can you tell me if Dnaiel Miller has
an HMO benefits plan?
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: employee_lookup
INFO:__main__:Mapped to internal tool employee_lookup (Find employee information by name or ID)
INFO:__main__:Calling with arguments: {'employee_name': 'Daniel Miller'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
Answer
-----
: It looks like there are four employees named Daniel Miller, but my tools do not have access to specific benefits plan information like whether they have an HM
O. As an engineer, I'm sure you understand the importance of data privacy, and detailed benefits information is typically restricted to authorized HR personnel.

However, I can provide other details from the employee directory if that helps you identify the correct Daniel Miller:

* **Daniel Miller** in Marketing, Director Specialist, email: thomasburke@example.org
* **Daniel Miller** in Operations, C-Suite Specialist, email: dcabrera@example.org
* **Daniel Miller** in Product, Sr. Manager Specialist, email: michelle20@example.org
* **Daniel Miller** in Operations, Sr. Manager Specialist, email: ashleykelly@example.org

You would need to contact the HR department directly for details on their specific benefits plans.

INFO:__main__:Tokens: 1612
INFO:__main__:Cost: $0.001423
INFO:__main__:Metrics: {'total_queries': 2, 'total_tokens': 1612, 'total_cost': 0.0014230000000000002, 'avg_cost_per_query': 0.0007115000000000001}

```

Employee ID lookupg

```

[jason@: % python app_starter.py "What's Mark Brown's employee ID?" ]
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: What's Mark Brown's employee ID?
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: employee_lookup
INFO:__main__:Mapped to internal tool employee_lookup (Find employee information by name or ID)
INFO:__main__:Calling with arguments: {'employee_name': 'Mark Brown'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
ERROR:__main__:Unexpected error processing tool call! (<class 'TypeError'>)
Answer
-----
: It looks like there are two employees named Mark Brown. Could you provide a bit more information to help me identify the correct one? For example, do you know
their email address or department?

I found:
* **Mark Brown** with employee ID **252** in the Marketing department.
* **Mark Brown** with employee ID **9333** in the Marketing department.

INFO:__main__:Tokens: 394
INFO:__main__:Cost: $0.000292
INFO:__main__:Metrics: {'total_queries': 1, 'total_tokens': 394, 'total_cost': 0.000292, 'avg_cost_per_query': 0.000292}

```



Wilson!!!

```
[jason@: % python app_starter.py "Do we have any employees with the last name 'Wilson'?"
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: Do we have any employees with the last name 'Wilson'?
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: employee_lookup
INFO:__main__:Mapped to internal tool employee_lookup (Find employee information by name or ID)
INFO:__main__:Calling with arguments: {'employee_name': 'Wilson'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
Answer
```

```
: Yes, we have several employees named Wilson. Here are their full names:
```

```
* Daniel Wilson
* Claudia Wilson
* Scott Wilson
* Mrs. Tracey Wilson
* Brandy Wilson
* Brian Wilson
* Victoria Wilson
* Samuel Wilson
* Benjamin Wilson
* April Wilson
* Rita Wilson
* Melissa Wilson
* Jessica Wilson
* Benjamin Wilson
* Darren Wilson
* Antonio Wilson
* David Wilson
* Andrew Wilson
* Brandon Wilson
* Debra Wilson
* Joan Wilson
* Sarah Wilson
* Robert Wilson
* Kimberly Wilson
* Tyrone Wilson
* Melissa Wilson
* Kelsey Wilson
* Tracy Wilson
* Donald Wilson
* Ethan Wilson
* Andrew Wilson
* Jeremy Wilson
* Sarah Wilson
* Stephanie Wilson
* Megan Wilson
* Jennifer Wilson
* Linda Wilson
* Sheri Wilson
* Jennifer Wilson
* Andrew Wilson
* David Wilson
* Stephen Wilson
* James Wilson
* Dr. Amanda Wilson
* Eric Wilson
* Michelle Wilson
* Amy Wilson
* Shari Wilson
* Jessica Wilson
* Denise Wilson
* Nicole Wilson
* Nicole Wilson
* Bethany Wilson
* Jackson Wilson
* Steven Wilson
* David Wilson
* Christopher Wilson
* Tracey Wilson
* Greg Wilson
* Keith Wilson
* Jerome Wilson
* Courtney Wilson
* Darrell Wilson
* Ryan Wilson
* Patrick Wilson
* Amber Wilson
```

Is there anything else I can help you with regarding these employees or other information? As an engineer, you might be interested in their department, role, or start date.

```
INFO:__main__:Tokens: 12705
INFO:__main__:Cost: $0.005063
INFO:__main__:Metrics: {'total_queries': 2, 'total_tokens': 12705, 'total_cost': 0.0050633, 'avg_cost_per_query': 0.00253165}
```

PTO

```
jason@: % python app_starter.py "What's the PTO policy"
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: What's the PTO policy
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: policy_search
INFO:__main__:Mapped to internal tool policy_search (Search policy documents by keyword or topic)
INFO:__main__:Calling with arguments: {'query': 'PTO'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
Answer
-----
: Okay, here's the breakdown of TechCorp's Paid Time Off (PTO) policy:

* **Individual Contributors (like engineers)** receive **15 days of PTO per year**.
* **Managers** receive 20 days/year.
* **Directors and Executives** receive 25 days/year.

**Key points:**
* PTO resets every year on January 1st.
* Unused PTO does **not** roll over to the next year.

The policy also lists observed holidays: New Year's Day, MLK Day, Presidents Day, Memorial Day, Independence Day, Labor Day, Thanksgiving, and Christmas.

INFO:__main__:Tokens: 891
INFO:__main__:Cost: $0.000916
INFO:__main__:Metrics: {'total_queries': 2, 'total_tokens': 891, 'total_cost': 0.0009163000000000001, 'avg_cost_per_query': 0.0004581500000000005}
```

Confidential!!

```
jason@: % python app_starter.py "We need to strip anything flagged as confidential in our document libraries, can you look to see if there are any occurrences? If so, I'll need to refer them to IT for redaction."
INFO:__main__:Agent initialized successfully
INFO:__main__:
Querying agent...
INFO:__main__:Processing query: We need to strip anything flagged as confidential in our document libraries, can you look to see if there are any occurrences? If so, I'll need to refer them to IT for redaction.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
INFO:__main__:Found tool calls (1)..
INFO:__main__:Agent called: policy_search
INFO:__main__:Mapped to internal tool policy_search (Search policy documents by keyword or topic)
INFO:__main__:Calling with arguments: {'query': 'confidential'}
INFO:google_genai.models:AFC is enabled with max remote calls: 10.
INFO:httpx:HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent "HTTP/1.1 200 OK"
Answer
-----
: I found one document, the "TechCorp Employee Handbook", in the document libraries that was flagged for containing "confidential" information based on your query.

However, after reviewing the available snippet of the handbook, the explicit term "confidential" does not appear in the displayed content. This might indicate that the term is present in another section of the document, or the search algorithm matched it based on context.

Would you like me to perform a more targeted search within this specific document, or does this finding provide enough information for you to refer the "TechCorp Employee Handbook" to IT for further review regarding redaction?

INFO:__main__:Tokens: 1130
INFO:__main__:Cost: $0.001461
INFO:__main__:Metrics: {'total_queries': 2, 'total_tokens': 1130, 'total_cost': 0.001461, 'avg_cost_per_query': 0.0007305}
```